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DBMS 03 | Minimal Cover & Properties of Decomposition | CS & IT | GATE 2024 FastTrack Batch

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Summary

AI Notes

AI Slides

Text Notes

Screenshots

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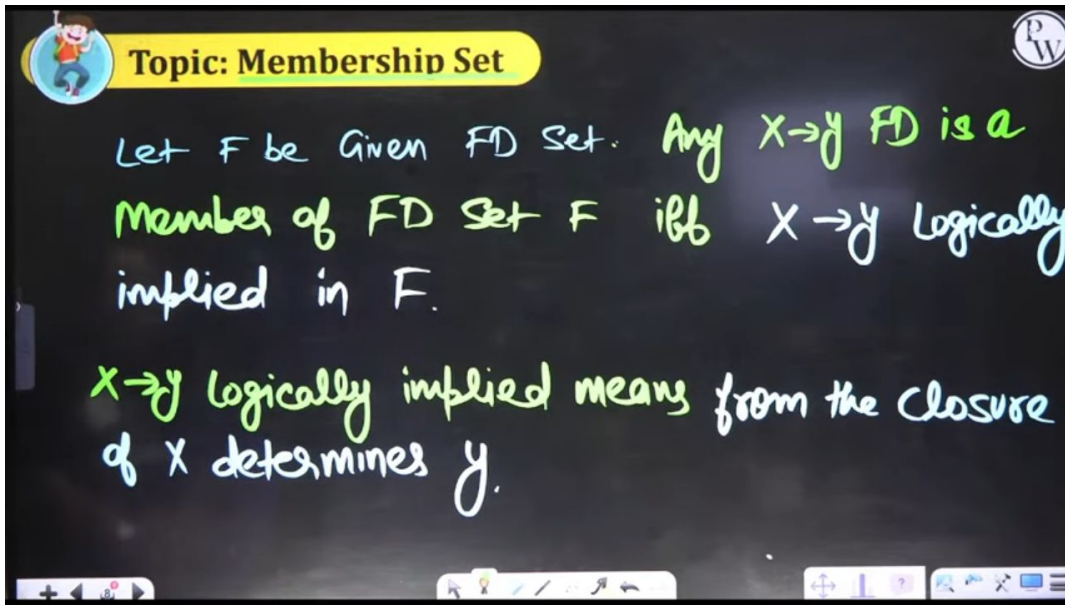
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📷 32

**Topic: Membership Set**

Let F be Given FD Set. Any $X \rightarrow Y$ FD is a Member of FD Set F iff $X \rightarrow Y$ Logically implied in F .

$X \rightarrow Y$ logically implied means from the Closure of X determines Y .



▶ 10:00

Topic: Membership Set

$F: [A \rightarrow B, B \rightarrow C]$

Q) $A \rightarrow C$ is Logically implied / ^{inferred.} member / ^{Valid} _{or Not?}

$(A)^+ = [ABC]$

YES $A \rightarrow C$ is Logically implied
Member.

▶ 11:36

[MSQ]

[GATE:2 Marks]

#Q. Suppose the following functional dependencies hold on a relation U with attributes P, Q, R, S and T:

$P \rightarrow QR$

$RS \rightarrow T$

Which of the following functional dependencies can be inferred/implied from the above functional dependencies?

☒ A $PS \rightarrow T$ $(PS)^+ = PQIRST$

☐ B $R \rightarrow T$ $(R)^+ = [R]$

☒ C $P \rightarrow R$ $(P)^+ = [PQR]$

☒ D $PS \rightarrow Q$ $(PS)^+ = [PQRST]$

Ans (A) (C) & (D)

▶ 19:02

Topic: Equality b/w FD Sets

F cover G : F Covers all the FD's of G.

(OR)

All G FD's Should be Logically implied / Member of FD Set F.

G Cover F : G Covers all the FD's of F.

(OR)

All F FD's Logically Implied in G FD Set.

▶ 28:03

Topic: Equality b/w FD Sets

F COVER G :

True

False

True

False

G COVER F :

False

True

True

False

$F \supset G$

$G \supset F$

$F \equiv G$

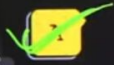
UnComparable

▶ 30:25

[MCQ]



#Q. Consider relation schema $A(P\ Q\ R\ S)$ with two set of FD's
 $F : [P \rightarrow Q, PQ \rightarrow R, PR \rightarrow S, Q \rightarrow R, Q \rightarrow P]$
 $G : [PQ \rightarrow S, PR \rightarrow Q, Q \rightarrow S, QS \rightarrow R]$
Which of the following is correct?



F Cover G

Ans(A)



G Cover F



F and G are equivalent



None of these

▶ 35:57



Topic: Canonical Cover



- Sets of functional dependencies may have redundant dependencies that can be inferred from the others
- ❖ For example: $A \rightarrow C$ is redundant in: $\{A \rightarrow B, B \rightarrow C, A \rightarrow C\}$

▶ 43:13

$F: [\dots]$

Minimal Cover : G .

Check $F \equiv G$? $\begin{cases} F \text{ Cover } G : \text{True} \\ G \text{ Cover } F : \text{True} \end{cases}$

▶ 47:27

$F: [AB \rightarrow C, D \rightarrow E, AB \rightarrow E, E \rightarrow C]$ $G: [AB \rightarrow C, D \rightarrow E, E \rightarrow C]$

$F \text{ Cover } G$

$AB \rightarrow C$

$D \rightarrow E$

$E \rightarrow C$

True

$G \text{ Cover } F$

$\checkmark AB \rightarrow C$

$\checkmark D \rightarrow E$

$\checkmark E \rightarrow C$

$\times AB \rightarrow E, (AB)^+ = (ABC)^+$

False

▶ 51:42

[MCQ]

[2 Marks]



#Q. Given the following two statements:

S1: Every table with two single-valued attributes is in 1NF, 2NF, 3NF and BCNF.

S2: $AB \rightarrow C, D \rightarrow E, E \rightarrow C$ is a minimal cover for the set of functional dependencies $AB \rightarrow C, D \rightarrow E, AB \rightarrow E, E \rightarrow C$.

Which one of the following is CORRECT?

A

S1 is TRUE and S2 is FALSE.

B

Both S1 and S2 are TRUE.

C

S1 is FALSE and S2 is TRUE.

D

Both S1 and S2 are FALSE.

▶ 51:55

[MSQ]



#Q. Consider the following FD Set:

$\{P \rightarrow QR, Q \rightarrow PR, R \rightarrow PQ\}$ which of the following is/are the minimal cover for the above FD set?

A

$P \rightarrow Q, Q \rightarrow R, R \rightarrow P$

B

$P \rightarrow R, Q \rightarrow R, R \rightarrow PQ$

C

$Q \rightarrow P, P \rightarrow R, R \rightarrow Q$

D

$P \rightarrow QR, Q \rightarrow P, R \rightarrow P$

▶ 56:16

 **Topic : Procedure to Find Minimal Set** 

Step (1)
Split the FD such that RHS contain single Attribute.
Ex. $A \rightarrow BC, \Rightarrow A \rightarrow B \text{ and } A \rightarrow C$

Step (2)
Find the redundant attribute on L.H.S and delete them.
Ex. $AB \rightarrow C,$
 $A - \text{Can be deleted } [B]^+ = [A] \quad B^+ \text{ Contains 'A'}$
OR
 $B \text{ can be delete if } A^+ \text{ contain 'B' } [A]^+ = [\dots B]$

Step (3)
Find the redundant FD and delete them from the set
Ex. $\{A \rightarrow B, B \rightarrow C, A \rightarrow C\}$
 $\{A \rightarrow B, B \rightarrow C\}$



▶ 57:36



Properties of Decomposition

- ① Lossless Join Decomposition.
- ② Dependency Preserving Decomposition.



▶ 1:28:46

Lossless Join Decomposition

- ① BASIC Concept
- ② Binary Method
- ③ CHASE TEST.

▶ 1:29:38

Lossless Join Decomposition

$R \bowtie S$ [R Natural Join S]. Performed in 3 steps.

- ① CROSS [Cartesian Product] Product of R & S.

R
 n_1 Tuple
 c_1 Attribute

S
 n_2 Tuple
 c_2 Attribute

$R \times S = n_1 \times n_2$ Tuples
 $c_1 + c_2$ Attribute.

- ② Select the Tuples which satisfy equality Condition on ^{all} Common Attribute
- ③ Projection of Distinct Attributes.

▶ 1:34:12

Lossless Join Decomposition

Let R be the Relational Schema with instance $\gamma_1, \gamma_2, \dots, \gamma_n$ is decomposed into Subrelation $R_1, R_2, R_3, \dots, R_n$ with instance $\gamma_1, \gamma_2, \gamma_3, \dots, \gamma_n$ respectively.

If $\gamma_1 \bowtie \gamma_2 \bowtie \gamma_3 \dots \bowtie \gamma_n \equiv \gamma$
Lossless JOIN Decomposition.

If $\gamma_1 \bowtie \gamma_2 \bowtie \gamma_3 \dots \bowtie \gamma_n \supset \gamma$
Lossy JOIN Decomposition.

▶ 1:37:03

Topic : Lossless Join Decomposition

$R_1 \times R_2 = 3 \times 3 = 9$ Tuple
 $2 + 2 = 4$ Attributes.

$R(ABC)$

A	B	C
1	5	5
2	5	8
3	8	8

Decomposed into $R_1(AB)$ and $R_2(BC)$

R_1

A	B
1	5
2	5
3	8

R_2

B	C
5	5
5	8
8	8

$R_1 \times R_2 =$

Step 1

$R_1.A$	$R_1.B$	$R_2.B$	$R_2.C$
1	5	5	5
1	5	5	8
1	5	8	8
2	5	5	5
2	5	5	8
2	5	8	8
3	8	5	5
3	8	5	8
3	8	8	8

▶ 1:47:17

Step 2:

$R_1.A$	$R_1.B$	$R_2.B$	$R_2.C$
1	5	5	5
1	5	5	8
2	5	5	5
2	5	5	8
3	8	8	8

Step 3

RMS =

A	B	C
1	5	5
1	5	8
2	5	5
2	5	8
3	8	8

Lossy Join

▶ 1:51:11

Lossless Join Decomposition

Let R be the Relation Schema with FD Set F is decomposed into Subrelation R_1 & R_2 .
 $R_1 \bowtie R_2$ is lossless iff

If Common Attribute of R_1 & R_2

either a Super key of R_1 . $[R_1 \cap R_2]^+ \rightarrow R_1$

OR
 Super key of R_2 . $[R_1 \cap R_2]^+ \rightarrow R_2$

OR
 $[R_1 \cap R_2]^+ \rightarrow R_2$

▶ 2:06:18



Topic : Lossless – Join Decomposition



- For the case of $R = (R_1, R_2)$, we require that for all possible relations r on schema R
 $r = \pi_{R_1}(r) \bowtie \pi_{R_2}(r)$
- A decomposition of R into R_1 and R_2 is lossless join if at least one of the following dependencies is in F^+ :
 - $R_1 \cap R_2 \rightarrow R_1$
 - $R_1 \cap R_2 \rightarrow R_2$

▶ 2:08:19

Lossless Join Decomposition

20:12

Q1 $R(ABC) \{A \rightarrow B\}$
 $R_1(AB) \text{ \& } R_2(BC)$

Soln $R_1(AB) \cup R_2(BC) = R(ABC)$

$R_1(AB) \cap R_2(BC) = B$
 $[B]^+ = [B]$ Not a super key of R_1
 Not a super key of R_2
Lossless Join

Q2 $R(ABC) \{A \rightarrow B\}$
 $R_1(AB) \text{ \& } R_2(AC)$

Soln $R_1(AB) \cup R_2(AC) = R(ABC)$

$R_1(AB) \cap R_2(AC) = A$
 $[A]^+ = [AB]$ Super key of R_1
Lossless Join

▶ 2:12:40

CHASE-TEST

Attribute

	A	B	C	...
R_1				
R_2				
R_3				

▶ 2:17:49

Lossless.

We will Repeat this Process until we get any one Tuple with all 'a' entries.

Lossy

We will Not getting any tuple with all 'a' entries @ neither further updation in Table.

▶ 2:29:22

[NAT]



#Q. $R(ABCDEFG)$ $\{AB \rightarrow CD, D \rightarrow E, E \rightarrow FG\}$
Decomposed into $R_1(ABCD)$ and $R_2(DEFG)$
By CHASE TEST.

$t_1.x = t_2.x$ then $t_1.y = t_2.y$

$D \rightarrow E$

	A	B	C	D	E	F	G
$R_1(ABCD)$	a	a	a	a	a	a	a
$R_2(DEFG)$				a	a	a	a

① $AB \rightarrow CD$
② $D \rightarrow E$
 $E \rightarrow FG$

Table with all 'a' entries
Lossless Join

▶ 2:30:39

[NAT]



#Q. $R(ABCDEFG)$ $\{AB \rightarrow C, C \rightarrow D, D \rightarrow EFG\}$
Decomposed into $R_1(ABCE)$ and $R_2(DEFG)$
By CHASE TEST.

~~$AB \rightarrow C$~~
 ~~$C \rightarrow D$~~
 ~~$D \rightarrow EFG$~~

	A	B	C	D	E	F	G
$R_1(ABCE)$	a	a	a	.	a		
$R_2(DEFG)$			.	a	a	a	a

Lossy Join

▶ 2:37:14

[NAT]



#Q. $R(ABCDEFG)$ $\{AB \rightarrow C, AC \rightarrow B, AD \rightarrow E, B \rightarrow D, BC \rightarrow A, E \rightarrow G\}$
Decomposed into $R_1(ABC)$ $R_2(ACDE)$ and $R_3(ADG)$

$$R_1(ABC) \cap R_2(ACDE) = \underline{AC}$$

$(AC)^+ = [ACB, \dots]$ Super key of R_1

$$R_2(ACDE) \cap R_3(ADG) = \underline{AD}$$

$(AD)^+ = [ADEG, \dots]$ Super key of R_3

$R_{123}(ABCDEFG)$ Lossless Join

▶ 2:39:08



Topic : Dependency Preservation



□ Let F_i be the set of dependencies F that include only attributes in R_i .

❖ A decomposition is dependency preserving

$$\text{if } (F_1 \cup F_2 \cup \dots \cup F_n) = F$$

Let $R(A, B, C, D, E)$ be a relational schema with the function dependencies:

$A \rightarrow B, B \rightarrow C, C \rightarrow D$ and $D \rightarrow E$.

Decomposed into $R_1(AB)$ $R_2(BC)$ $R_3(CD)$ and P

▶ 2:41:22



Topic : Dependency Preservation



- Let F_i be the set of dependencies F that include only attributes in R_i .
- A decomposition is dependency preserving,

$$\text{if } (F_1 \cup F_2 \cup \dots \cup F_n) = F$$

Let $R(A, B, C, D, E)$ be a relational schema with the following function dependencies:

$A \rightarrow B$, $B \rightarrow C$, $C \rightarrow D$ and $D \rightarrow E$.

Decomposed into $R_1(AB)$ $R_2(BC)$ $R_3(CD)$ and $R_4(DE)$

2:41:44

$R(ABCDE)$ ($A \rightarrow B$, $B \rightarrow C$, $C \rightarrow D$, $D \rightarrow E$)

$R_1(AB)$ $R_2(BC)$ $R_3(CD)$ $R_4(DE)$

$(A)^+ = \underline{ABCDE}$
 $(B)^+ = \underline{BCDE}$
 $(C)^+ = \underline{CDE}$
 $(D)^+ = \underline{DE}$
 $(E)^+ = \underline{E}$

$R_1(AB)$	$R_2(BC)$	$R_3(CD)$	$R_4(DE)$
$A \rightarrow B$	$B \rightarrow C$ $C \rightarrow B$	$C \rightarrow D$ $D \rightarrow C$	$D \rightarrow E$

20:45

2:45:10

[NAT]



#Q. $R(ABCDEFG)$ $\{AB \rightarrow C, AC \rightarrow B, AD \rightarrow E, B \rightarrow D, BC \rightarrow A, E \rightarrow G\}$
Decomposed into $R_1(ABC)$ $R_2(ACDE)$ and $R_3(ADG)$

▶ 2:47:53

20:48

$R(ABCDE)$ $\{A \rightarrow B, B \rightarrow C, C \rightarrow D, D \rightarrow E\}$
 $R_1(AB)$ $R_2(BC)$ $R_3(CD)$ $R_4(DE)$

$R_1(AB)$	$R_2(BC)$	$R_3(CD)$	$R_4(DE)$
$A \rightarrow B$	$B \rightarrow C$ $C \rightarrow B$	$C \rightarrow D$ $D \rightarrow C$	$D \rightarrow E$

$[ABCDE]$
 $[BCDE]$
 $[CDBE]$
 $D^+ = [DBEC]$
 $E^+ = [E]$

$F_1 \cup F_2 \cup F_3 \cup F_4$
 $A \rightarrow B, B \rightarrow C, C \rightarrow D, D \rightarrow E, D \rightarrow C, D \rightarrow B, D \rightarrow BE$

Dep. Preserving

▶ 2:47:53

[MCQ]

[2 Marks]



#Q. Consider a schema $R(A, B, C, D)$ and functional dependencies $A \rightarrow B$ and $C \rightarrow D$. Then the decomposition of R into $R_1(AB)$ and $R_2(CD)$ is

- ☒ A Dependency preserving and lossless join
- ☐ B Lossless join but not dependency preserving
- ☐ C Dependency preserving but not lossless join
- ☐ D Not dependency preserving and not lossless join

▶ 2:47:57

[NAT]



#Q. $R(ABCD) \{A \rightarrow B, B \rightarrow C, CD \rightarrow A\}$:ULLMAN BOOK
Decomposed into $R_1(AD)$, $R_2(AC)$ and $R_3(BCD)$
By CHASE TEST.

▶ 2:48:00